

Wanglin He

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EDUCATION

Columbia University

MS in Mechanical Engineering

- **GPA:** 3.6/4.0

- **Core Courses:** Robotic Studio(Advanced); Robot Learning; Introduction to Robotics

New York, US

Sep 2025 – Current

Shanghai University of Engineering Science

B.Eng. in Mechanical Engineering

- **GPA:** 3.8/4.0

- **Honors:** Outstanding Graduate of Shanghai; 2nd Prize & 3rd Prize of SUES Outstanding Student Scholarship

- **Core Courses:** Intro Robotics; Mechanical Design; Electronic Technique; Electrical Technology; Engineering Mechanics; Mechanical Engineering and Automation Cognition Practice

Shanghai, CN

Sep 2020 – Jun 2024

PUBLICATION

Wanglin He, Siyuan Zhang, Junyan liu. Robotic Budding: From Homogeneous Cells to Diverse Morphologies in Physical Self-Reproduction. <https://openreview.net/forum?id=2dITymceecM&referrer=%5BAuthor>

Sun, C., **He, W.**, Wen, H. (2023). Mechanical training arm to assist in fracture rehabilitation. *International Journal of Technical & Scientific Research Engineering*, 6(3). <https://www.ijtsre.org/papers/2023/ev6c3/IJT-44712288.pdf>.

RESEARCH PROJECT

A Robustness Benchmark for Vision-Language-Action Models under Visual Occlusion and Long-Horizon Cyclic Manipulation

May 2026 – Current

Collaborative research with researchers at MIT and HKUST

- Developed an Isaac Sim-based benchmark for systematically evaluating the robustness of Vision-Language-Action (VLA) models under visual occlusion and long-horizon cyclic manipulation tasks.
- Integrated and evaluated three state-of-the-art VLA models—X-VLA, $\pi 0.5$, and G0.5—within a unified simulation pipeline, standardizing visual observations, language instructions, action interfaces, control frequencies, environment resets, and task-success criteria to enable fair comparisons.
- Designed controlled evaluation protocols that separate baseline manipulation ability from performance degradation caused by occlusion, extended task horizons, and repeated task cycles, enabling analysis of compounding errors and post-occlusion recovery behavior.
- Established a failure-analysis framework to identify perception loss, task-progress confusion, repeated actions, and motion drift, providing the experimental foundation for developing model-independent memory and recovery mechanisms.

Self-Reproducing Robot Project: Robotic Control Optimization Simulation

Sep 2025 – Current

Creative Machine Lab Supervised by Prof. Hod Lipson,

- Spearheaded the strategic migration of the simulation framework to DeepMind's MuJoCo/MJX, configuring high performance cloud servers (vCPU) and local Windows GPU acceleration to optimize large-scale evaluations for random search and genetic algorithms.
- Engineered a low-level magnetic physics engine to overcome MuJoCo's lack of native complex magnetic support, formulating an $O(N^2)$ magnetic calculation model featuring non-linear distance attenuation and velocity damping.
- Overcame Python's matrix processing bottlenecks by developing a highly optimized C dynamic library. Bypassed the Python interpreter to directly inject force data into MuJoCo's low-level array, achieving a 2.34x speedup per simulation step and drastically accelerating the physical evaluation process.
- Diagnosed and resolved low-level C compilation and linking issues to maintain system stability; systematically tracked optimal hyperparameters and fitness scores to comprehensively evaluate algorithmic search performance.

CONTESTS EXPERIENCES

RoboMaster University Competition Series

Mar 2022 – Jun 2024

Mechanical Engineer & Hardware Engineer | Supervised by Prof. Chunyan Zhang, SUES

- Led the mechanical structure design, circuit board drawing, and wiring layout utilizing Solidworks and Altium Designer of the heroic robot

- Categorized the robot structures into turret mechanism and chassis structure, ensuring smooth operation of the firing mechanism and unrestricted movement along the pitch and yaw axes
- Developed the chassis with mecanum wheels and multi-stage damping devices to facilitate omni-directional movement capabilities and utilized a collector slip ring to connect the turret and chassis, enabling independent motion without interference with the circuitry of the turret

TEACHING EXPERIENCE

Graduate Teaching Assistant | Columbia University

Sep 2026 – Current

MECE 4611 Robotic Studio

- Contributed to robotics teaching and curriculum development, integrating python algorithm/ embedded systems to support 120+ students in hands-on robotic projects across undergraduate and graduate levels.
- Provided technical support and debugging help while maintaining positive student feedback across course evaluations.

SKILLS

- **Computer:** Solidworks, Python, Catia, Altium Designer, Matlab, C